

# Companion to Bird & de Moor, Algebra of Programming

## §6. Recursive Programs

### §6.1. Digits of a number

(6.1a)

$$([g,h])^\circ = (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ) \text{ snoc}$$

the converse of a fold over decimals undoes the algebra  $[g,h]$  and then either **wrap**, or the fold's converse on the front and **snoc**

|   |                                                                                                       |                                                                                                             |
|---|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1 | $[g,h]^\circ$                                                                                         | $(\emptyset) = [\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+-}(\neg \text{Digit}(\emptyset)) \emptyset$ |
| 2 | $= ([\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+-}(\neg \text{Digit}([g,h])^\circ)) [g,h]^\circ$ | $(R + S) [P,Q] = [R P, S Q]$                                                                                |
| 3 | $= ([\text{wrap}, \text{snoc}]^\circ [g, \neg \text{Digit}([g,h])^\circ h])^\circ$                    | $[U,V]^\circ [R,S] = (U^\circ R) \cup (V^\circ S)$                                                          |
| 4 | $= ((\text{wrap}^\circ g) \cup (\text{snoc}^\circ \neg \text{Digit}([g,h])^\circ h))^\circ$           | $(R \cup S)^\circ = S^\circ \cup R^\circ$                                                                   |
| 5 | $= (\text{snoc}^\circ \neg \text{Digit}([g,h])^\circ h)^\circ \cup (\text{wrap}^\circ g)^\circ$       | $R \cup S = S \cup R$                                                                                       |
| 6 | $= (\text{wrap}^\circ g)^\circ \cup (\text{snoc}^\circ \neg \text{Digit}([g,h])^\circ h)^\circ$       | $(R S)^\circ = S^\circ R^\circ$                                                                             |
| 7 | $= (g^\circ \text{ wrap}) \cup ((\neg \text{Digit}([g,h])^\circ h)^\circ \text{ snoc})$               | $(R S)^\circ = S^\circ R^\circ$                                                                             |
| 8 | $= (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ \text{ snoc})$                 | $\neg \text{Digit}(R)^\circ = \neg \text{Digit}(R^\circ)$                                                   |
| 9 | $= (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ \text{ snoc})$                 |                                                                                                             |

$\text{val}^\circ = (\text{embed}^\circ \text{ wrap}) \cup (\text{op}^\circ \neg \text{Digit}(\text{val}^\circ) \text{ snoc})$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(6.1b)

- |   |                                                                                                                                                                                        |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\text{op}^\circ(m,p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$<br>$\text{op}^\circ$ splits a number into its quotient and remainder by 10                                       |
| 2 | $(\exists p. 0 < \pi_1(p) \wedge \text{op}^\circ(m,p)) \iff 10 \leq m$<br>$\text{op}^\circ$ gives a pair with a nonzero first component exactly at the numbers with two or more digits |
| 3 | $(\exists d. \text{embed}^\circ(m,d)) \iff 0 < m \wedge m < 10$<br>$\text{embed}^\circ$ gives a digit exactly at the one-digit numbers                                                 |

## §6.2. Least fixed points

(6.2a)

$\text{Monotonic}(\varphi) \triangleq \forall X. \forall Y. X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$  —  $\varphi$  maps the hom-set  $A \rightarrow B$  into itself with  $X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$ .

$(\mu X : \varphi(X))$  is the least  $X : A \rightarrow B$  with  $\varphi(X) \sqsubseteq X$ .

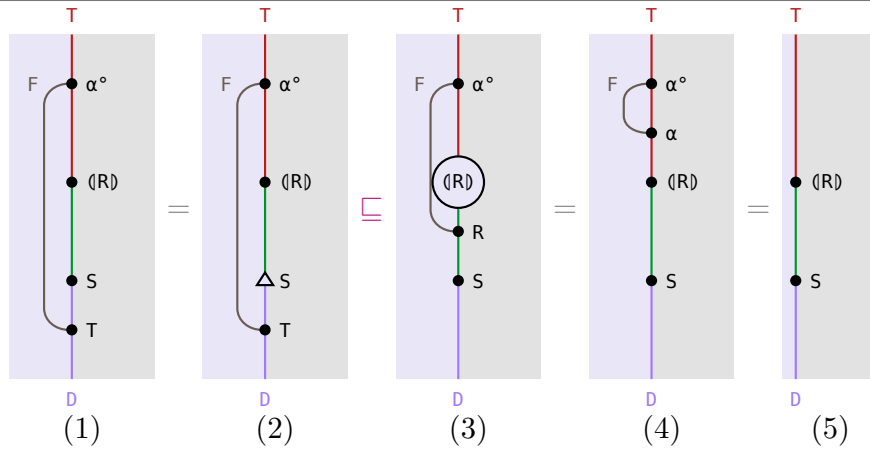
(6.2b)

| the law                                                                                   | what it says                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$                  | to bound $(\mu X : \varphi(X))$ above, exhibit one $Y$ the body does not grow past — the half §Section 6.3 and every chapter after it uses                                      |
| 2 $\text{Monotonic}(\varphi) \implies \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$ | <b>Knaster–Tarski</b> : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$ , so the least prefix point and the least fixed point are one relation |
| 3 $\alpha^\circ F(X) R \sqsubseteq X \implies \langle R \rangle \sqsubseteq X$<br>(6.2)   | a relation $X$ that contains one unfolding of the fold's recursion at $X$ contains the fold                                                                                     |
| 4 $X \sqsubseteq \alpha^\circ F(X) R \implies X \sqsubseteq \langle R \rangle$<br>(6.3)   | a relation $X$ contained in one unfolding of the fold's recursion at $X$ is contained in the fold                                                                               |

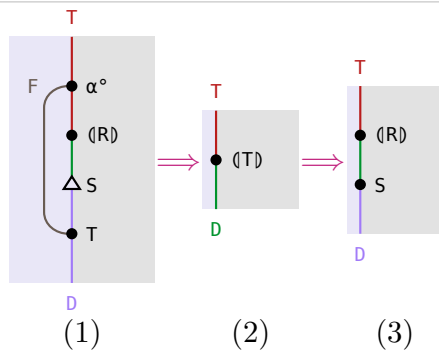
$$F(S) \ T \sqsubseteq R \ S \implies \langle T \rangle \sqsubseteq \langle R \rangle \ S$$

(6.2c)

if  $S$  followed by  $R$  absorbs  $T$  after  $F(S)$ , the fold of  $T$  is below the fold of  $R$  followed by  $S$



|                                                         |                                                         |
|---------------------------------------------------------|---------------------------------------------------------|
| 1 $\alpha^\circ F(\langle R \rangle S) T$               |                                                         |
| 2 $= \alpha^\circ F(\langle R \rangle) F(S) T$          | $F(S) \ T \sqsubseteq R \ S$                            |
| 3 $\sqsubseteq \alpha^\circ F(\langle R \rangle) R \ S$ | $\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$ |
| 4 $= \alpha^\circ \alpha \ \langle R \rangle \ S$       | $\alpha^\circ \alpha = 1$                               |
| 5 $= \langle R \rangle \ S$                             |                                                         |

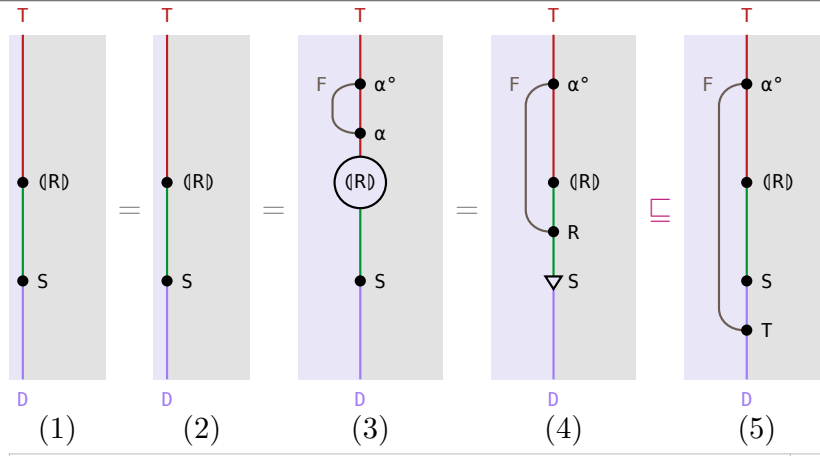


|                                                                                    |                                                                                                                  |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1 $F(S) \ T \sqsubseteq R \ S$                                                     | $F(S) \ T \sqsubseteq R \ S$<br>$\implies \alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S$ |
| 2 $\implies \alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S$ | $\alpha^\circ F(X) R \sqsubseteq X$<br>$\implies \langle R \rangle \sqsubseteq X$                                |
| 3 $\implies \langle T \rangle \sqsubseteq \langle R \rangle S$                     |                                                                                                                  |

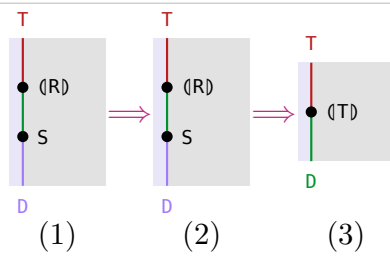
$$R \ S \sqsubseteq F(S) \ T \Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$$

(6.2d)

if  $R$  followed by  $S$  is absorbed by  $F(S)$  followed by  $T$ , the fold of  $R$  followed by  $S$  is below the fold of  $T$



|   |                                                           |                                                         |
|---|-----------------------------------------------------------|---------------------------------------------------------|
| 1 | $\langle R \rangle \ S$                                   |                                                         |
| 2 | $= \langle R \rangle \ S$                                 | $\alpha^\circ \ \alpha = 1$                             |
| 3 | $= \alpha^\circ \ \alpha \ \langle R \rangle \ S$         | $\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$ |
| 4 | $= \alpha^\circ \ F(\langle R \rangle) \ R \ S$           | $R \ S \sqsubseteq F(S) \ T$                            |
| 5 | $\sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$ |                                                         |



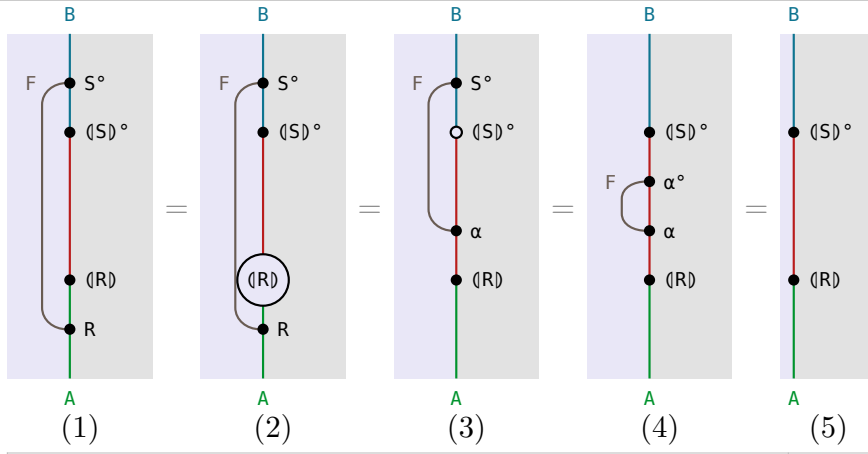
|   |                                                                                             |                                                                                                                             |
|---|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1 | $R \ S \sqsubseteq F(S) \ T$                                                                | $R \ S \sqsubseteq F(S) \ T$<br>$\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$ |
| 2 | $\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$ | $X \sqsubseteq \alpha^\circ \ F(X) \ R$<br>$\Rightarrow X \sqsubseteq \langle R \rangle$                                    |
| 3 | $\Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$                           |                                                                                                                             |

## §6.3. Hylomorphisms

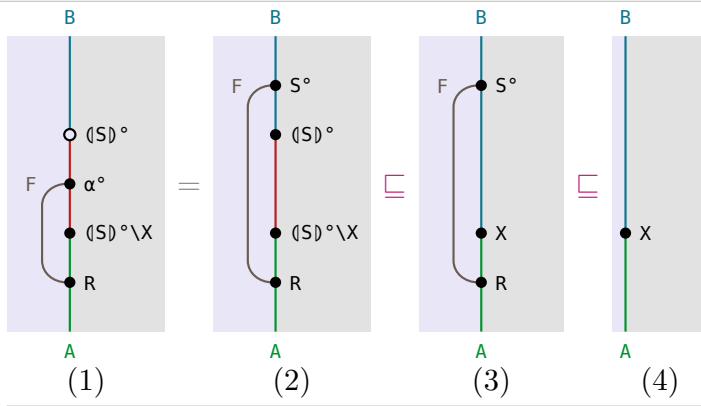
$$(S)^\circ (R) = (\mu X : S^\circ F(X) R)$$

(6.3a)

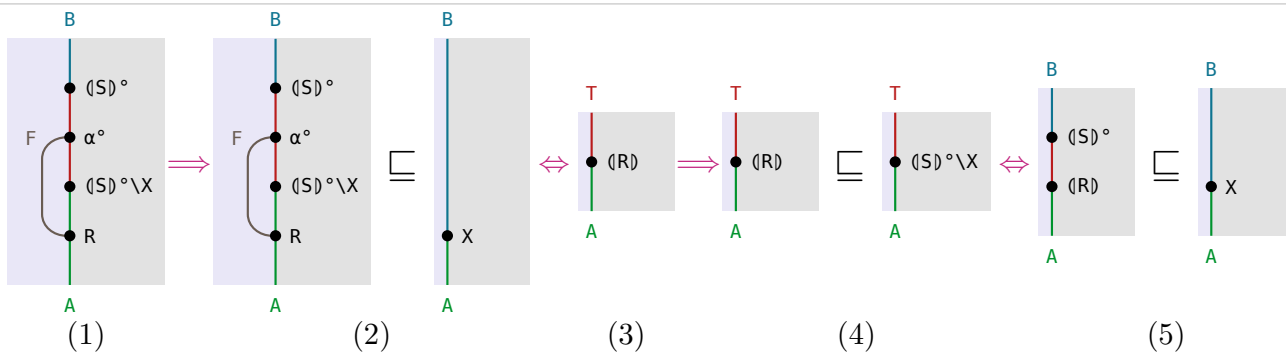
hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation



|   |                                       |                                                 |
|---|---------------------------------------|-------------------------------------------------|
| 1 | $S^\circ F((S)^\circ (R)) R$          |                                                 |
| 2 | $= S^\circ F((S)^\circ) F((R)) R$     | $\alpha (R) = F((R)) R$                         |
| 3 | $= S^\circ F((S)^\circ) \alpha (R)$   | $(R)^\circ \alpha^\circ = R^\circ F((R)^\circ)$ |
| 4 | $= (S)^\circ \alpha^\circ \alpha (R)$ | $\alpha^\circ \alpha = 1$                       |
| 5 | $= (S)^\circ (R)$                     |                                                 |



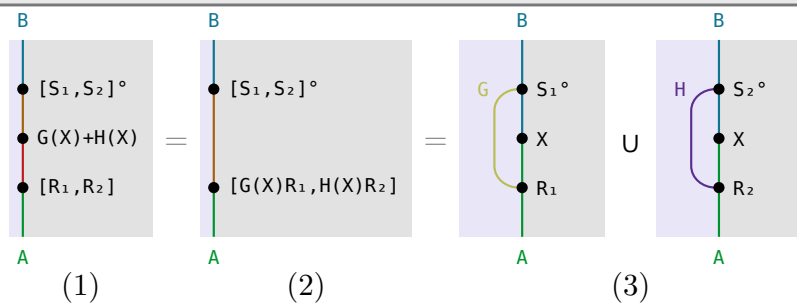
|   |                                                     |                                                 |
|---|-----------------------------------------------------|-------------------------------------------------|
| 1 | $(S)^\circ \alpha^\circ F((S)^\circ \setminus X) R$ | $(R)^\circ \alpha^\circ = R^\circ F((R)^\circ)$ |
| 2 | $= S^\circ F((S)^\circ ((S)^\circ \setminus X)) R$  | $S(S \setminus R) \subseteq R$                  |
| 3 | $\subseteq S^\circ F(X) R$                          | $S^\circ F(X) R \subseteq X$                    |
| 4 | $\subseteq X$                                       |                                                 |



|   |                                                                                           |                                                                             |
|---|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1 | $S^\circ F(X) R \subseteq X$                                                              | $S^\circ F(X) R \subseteq X$                                                |
| 2 | $\Rightarrow (S)^\circ \alpha^\circ F((S)^\circ \setminus X) R \subseteq X$               | $\Rightarrow (S)^\circ \alpha^\circ F((S)^\circ \setminus X) R \subseteq X$ |
| 3 | $\Leftrightarrow \alpha^\circ F((S)^\circ \setminus X) R \subseteq (S)^\circ \setminus X$ | $X \subseteq S \setminus R \Leftrightarrow S X \subseteq R$                 |
| 4 | $\Rightarrow (R) \subseteq (S)^\circ \setminus X$                                         | $\alpha^\circ F(X) R \subseteq X$                                           |
| 5 | $\Leftrightarrow (S)^\circ (R) \subseteq X$                                               | $\Rightarrow (R) \subseteq X$                                               |
|   |                                                                                           | $X \subseteq S \setminus R \Leftrightarrow S X \subseteq R$                 |

$$([S_1, S_2]^\circ)^\circ ([R_1, R_2])^\circ = (\mu X : (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results



|   |                                                    |                                                      |
|---|----------------------------------------------------|------------------------------------------------------|
| 1 | $[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$        | $(R + S) [P, Q] = [R P, S Q]$                        |
| 2 | $= [S_1, S_2]^\circ [G(X) R_1, H(X) R_2]$          | $[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$ |
| 3 | $= (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)$ |                                                      |

## §6.4. Fast exponentiation and modulus computation

| $\text{shift } f = (f \times 1) \ h \Rightarrow (\mu X : (\text{zero}^\circ \text{ zero } f) \cup (\text{shift}^\circ (X \times 1) h)) \sqsubseteq f$ <p>once <b>f</b> turns <b>shift</b> into <b>h</b>, the recursion that halves the argument at each step computes <b>f</b></p> |                                                                                          | (6.4a)                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-around; margin-top: 10px;"> <span>(1)</span> <span>(2)</span> <span>(3)</span> <span>(4)</span> </div>                                                                                                                           |                                                                                          |                                                                                                            |
| 1                                                                                                                                                                                                                                                                                  | $(\mu X : (\text{zero}^\circ \text{ zero } f) \cup (\text{shift}^\circ (X \times 1) h))$ | $\text{convert}^\circ ([g, h]) = (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$ |
| 2                                                                                                                                                                                                                                                                                  | $= \text{convert}^\circ ([\text{zero } f, h])$                                           | $\text{zero } f = g \wedge \text{shift } f = (f \times 1) \ h$                                             |
| 3                                                                                                                                                                                                                                                                                  | $= \text{convert}^\circ \text{ convert } f$                                              | $\Rightarrow \text{convert } f = ([g, h])$                                                                 |
| 4                                                                                                                                                                                                                                                                                  | $\sqsubseteq f$                                                                          | $\text{Simple}(\text{convert})$                                                                            |

| $(\mu X : (\text{zero}^\circ \text{ one}) \cup (\text{shift}^\circ (X \times 1) \text{ op}(a))) \sqsubseteq \text{exp}(a)$ <p>the recursion that halves the exponent at each step computes <b>a</b> to the power <b>b</b></p> |  | (6.4b) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|
| $\text{zero exp}(a) = \text{one}$<br>$a$ to the power 0 is 1                                                                                                                                                                  |  |        |
| $\text{shift exp}(a) = (\text{exp}(a) \times 1) \text{ op}(a)$<br>$a$ to the power $2n+d$ is $\text{op}(a)$ of $a$ to the power $n$ and $d$                                                                                   |  |        |

| $(\mu X : (\text{zero}^\circ \text{ zero}) \cup (\text{shift}^\circ (X \times 1) \text{ op}(b))) \sqsubseteq \text{mod}(b)$ <p>the recursion that halves <b>a</b> at each step computes the remainder of <b>a</b> by <b>b</b></p> |  | (6.4c) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|
| $\text{zero mod}(b) = \text{zero}$<br>$0 \text{ mod } b$ is 0                                                                                                                                                                     |  |        |
| $\text{shift mod}(b) = (\text{mod}(b) \times 1) \text{ op}(b)$<br>$(2a+d) \text{ mod } b$ is $\text{op}(b)$ of $a \text{ mod } b$ and $d$                                                                                         |  |        |

## §6.5. Unique fixed points

|                                                                                                                                                                                                                                                                                                                          |                               |                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------|
| $R R \sqsubseteq S R \Rightarrow (X / R / S) R R \sqsubseteq X$ $((X/R)/S) \text{ followed by } RR \text{ stays below } X \text{ when } RR \sqsubseteq SR$ $\text{inductive}(S) \wedge R R \sqsubseteq S R \Rightarrow \text{inductive}(R)$ $R \text{ is inductive when } RR \sqsubseteq SR \text{ for an inductive } S$ |                               | (6.5a)                    |
|                                                                                                                                                                                                                                                                                                                          |                               |                           |
| 1                                                                                                                                                                                                                                                                                                                        | $(X / R / S) R R$             | $R R \sqsubseteq S R$     |
| 2                                                                                                                                                                                                                                                                                                                        | $\sqsubseteq (X / R / S) S R$ | $(R / S) S \sqsubseteq R$ |
| 3                                                                                                                                                                                                                                                                                                                        | $\sqsubseteq (X / R) R$       | $(R / S) S \sqsubseteq R$ |
| 4                                                                                                                                                                                                                                                                                                                        | $\sqsubseteq X$               |                           |

## §6.5.1. Membership

|                                                                      |          |
|----------------------------------------------------------------------|----------|
| $\text{member} = \emptyset$                                          | (6.5.1a) |
| a constant relator records no elements, so its membership is empty   |          |
| $\text{member} = [\text{member}, \text{member}]$                     |          |
| a member of $F+G$ is a member of whichever summand the value is in   |          |
| $\text{member} = (\pi_1 \text{ member}) \cup (\pi_2 \text{ member})$ |          |
| a member of $F \times G$ is a member of either component             |          |

## §6.5.2. Consequences

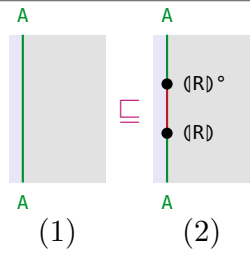
|                                                                                                                                                                                     |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| $\text{inductive}(S \text{ member}) \wedge X = S F(X) R \wedge Y = S F(Y) R \Rightarrow X = Y$ when $S \text{ member}$ is inductive, two solutions $X, Y$ of $X = SF(X)R$ are equal | (6.5.2a) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

|                                                                                                                                                                                                                                                               |          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| $\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S F(X) R \sqsubseteq X \Rightarrow \text{Entire}(X)$ when $S \text{ member}$ is inductive and $S, R$ are entire, a pre-fixed point of $X \mapsto SF(X)R$ is entire | (6.5.2b) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

|                                                                                                                                                                                                                                                                              |                                         |                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------|
| $\text{inductive}(g \text{ member}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = g F(X) f \Rightarrow \text{Map}(X)$ when $g \text{ member}$ is inductive and $f, g$ are maps, the solution of $X = gF(X)f$ is a map                                                 |                                         | (6.5.2c)                        |
| $X = g F(X) f \wedge \text{Simple}(g) \wedge \text{Simple}(f) \wedge Y^\circ X \sqsubseteq 1 \Rightarrow (g F(Y) f)^\circ X \sqsubseteq 1$ $(gF(Y)f)^\circ \text{ followed by } X \text{ stays below } 1 \text{ when } g, f \text{ are simple and } Y^\circ X \sqsubseteq 1$ |                                         |                                 |
|                                                                                                                                                                                                                                                                              |                                         |                                 |
| 1                                                                                                                                                                                                                                                                            | $(g F(Y) f)^\circ X$                    | $X = g F(X) f$                  |
| 2                                                                                                                                                                                                                                                                            | $= (g F(Y) f)^\circ g F(X) f$           | $(R S)^\circ = S^\circ R^\circ$ |
| 3                                                                                                                                                                                                                                                                            | $= f^\circ F(Y)^\circ g^\circ g F(X) f$ | $\text{PreservesRecip}(F)$      |
| 4                                                                                                                                                                                                                                                                            | $= f^\circ F(Y)^\circ g^\circ g F(X) f$ | $\text{Simple}(g)$              |
| 5                                                                                                                                                                                                                                                                            | $\sqsubseteq f^\circ F(Y)^\circ X f$    | $Y^\circ X \sqsubseteq 1$       |
| 6                                                                                                                                                                                                                                                                            | $\sqsubseteq f^\circ f$                 | $\text{Simple}(f)$              |
| 7                                                                                                                                                                                                                                                                            | $\sqsubseteq 1$                         |                                 |

(6.5.2d)

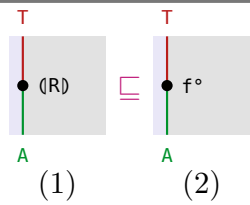
**$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R \Rightarrow 1 \sqsubseteq (R)^\circ (R)$**   
 when  $R^\circ \text{member}$  is inductive and  $R$  is surjective, the fold  $(R)$  is surjective



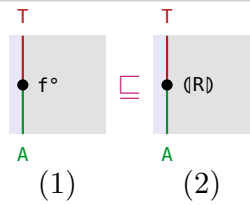
|   |                             |                                                                                                                                                                                 |
|---|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $1$                         | $\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S$<br>$F(X) R \sqsubseteq X$<br>$\Rightarrow \text{Entire}(X) \text{ at } S=R^\circ$ |
| 2 | $\sqsubseteq (R)^\circ (R)$ |                                                                                                                                                                                 |

(6.5.2e)

**$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = (R)$**   
 a map  $f$  with  $Rf \sqsubseteq F(f)\alpha$ , for a surjective  $R$ , has the fold  $(R)$  as its converse



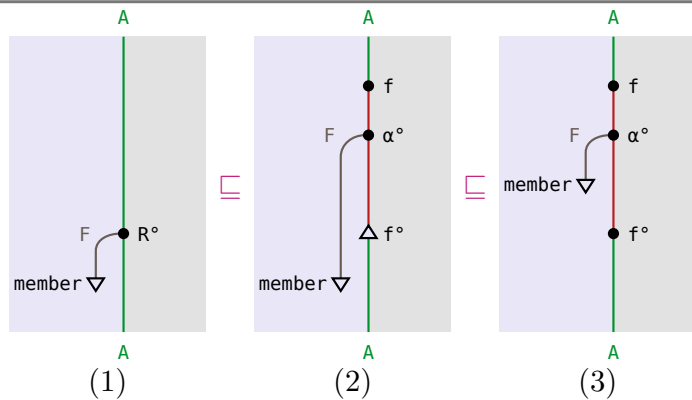
|   |                       |                                                    |
|---|-----------------------|----------------------------------------------------|
| 1 | $(R)$                 | shunting $f$ , fusion, $Rf \sqsubseteq F(f)\alpha$ |
| 2 | $\sqsubseteq f^\circ$ |                                                    |



|   |                   |                                                                                                                                                        |
|---|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $f^\circ$         | $\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R$<br>$\Rightarrow 1 \sqsubseteq (R)^\circ (R) \text{ and } (R)f \sqsubseteq 1$ |
| 2 | $\sqsubseteq (R)$ |                                                                                                                                                        |

(6.5.2f)

**$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \Rightarrow R^\circ \text{ member} \sqsubseteq f \alpha^\circ \text{ member } f^\circ$**   
 $R^\circ \text{member}$  is below  $\alpha^\circ \text{member}$  conjugated by  $f$ , so it is inductive when  $\alpha^\circ \text{member}$  is



|   |                                                        |                                                                                                                   |
|---|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 1 | $R^\circ \text{ member}$                               | $\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha$<br>$\Rightarrow R^\circ \sqsubseteq f \alpha^\circ F(f^\circ)$ |
| 2 | $\sqsubseteq f \alpha^\circ F(f^\circ) \text{ member}$ | $F(R) \text{ member} \sqsubseteq \text{member } R$                                                                |
| 3 | $\sqsubseteq f \alpha^\circ \text{ member } f^\circ$   |                                                                                                                   |

## §6.6. Sorting by selection

### §6.6.1. The function sort

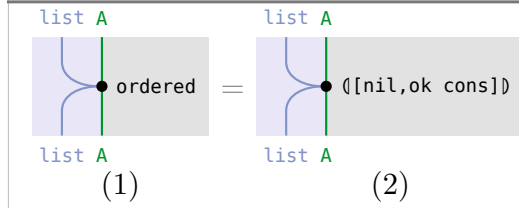
(6.6.1a)

| definition                                                                                                                       | type                                    | meaning                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------|
| 1 $\text{orderedP}[] = \text{true}, \text{orderedP}([a]\#xs) = (\forall x. x \in xs \implies R(a,x)) \wedge \text{orderedP}(xs)$ | $P[A]$                                  | every element is R-below each element after it                            |
| 2 $xs \text{ ordered } ys \iff xs = ys \wedge \text{orderedP}(xs)$                                                               | $[A] \rightarrow [A]$                   | the coreflexive passing exactly the lists <b>orderedP</b> holds of        |
| 3 $(a,xs) \text{ ok } q \iff (a,xs) = q \wedge (\forall x. x \in xs \implies R(a,x))$                                            | $A \times [A] \rightarrow A \times [A]$ | the coreflexive passing $(a, x)$ when $a$ is R-below every element of $x$ |

**ordered =  $\llbracket \text{nil}, \text{ok cons} \rrbracket$** 

(6.6.1b)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

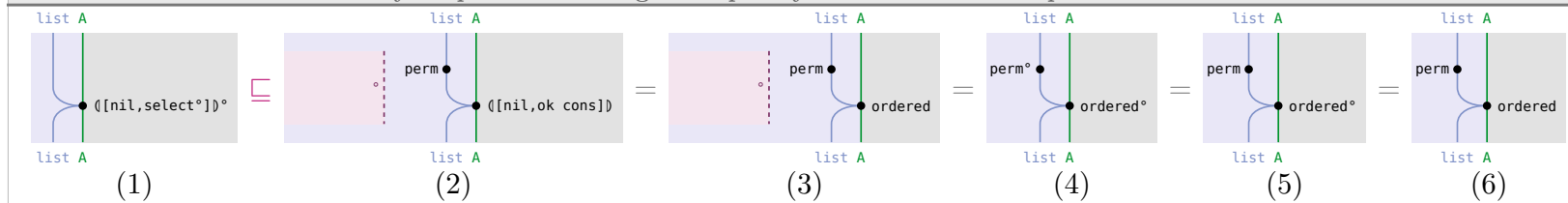


|                                                        |                 |
|--------------------------------------------------------|-----------------|
| 1 <b>ordered</b>                                       | fold uniqueness |
| 2 $= \llbracket \text{nil}, \text{ok cons} \rrbracket$ |                 |

### §6.6.2. Selection sort

 **$\text{select}^\circ \sqsubseteq \text{ok cons perm} \implies \llbracket \text{nil}, \text{select}^\circ \rrbracket^\circ \sqsubseteq \text{perm ordered}$** 

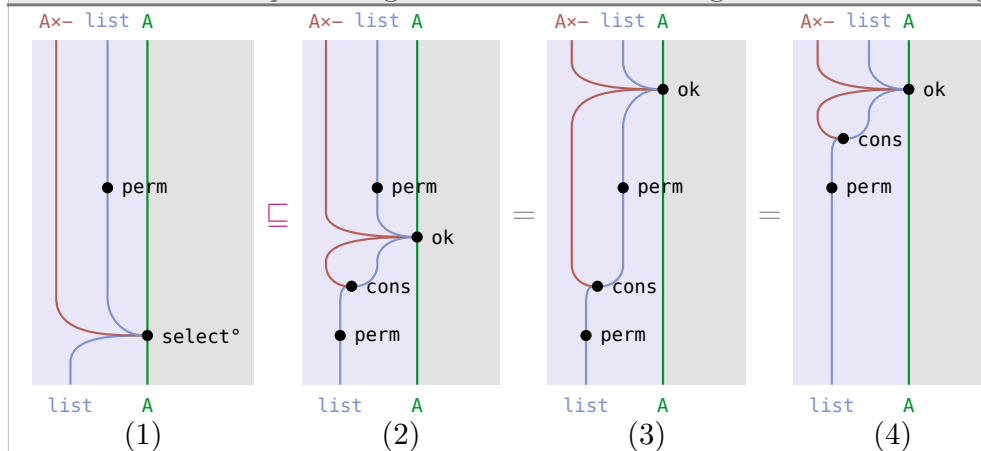
(6.6.2a)

every output of unfolding the input by **select** is a sorted permutation of it

|                                                                                       |                                                                                                                                                          |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $\llbracket \text{nil}, \text{select}^\circ \rrbracket^\circ$                       | $\text{select}^\circ \sqsubseteq \text{ok cons perm}$                                                                                                    |
| 2 $\sqsubseteq (\llbracket \text{nil}, \text{ok cons} \rrbracket \text{ perm})^\circ$ | $\implies \llbracket \text{nil}, \text{select}^\circ \rrbracket^\circ \sqsubseteq (\llbracket \text{nil}, \text{ok cons} \rrbracket \text{ perm})^\circ$ |
| 3 $= (\text{ordered perm})^\circ$                                                     | (6.6.1b) $\text{ordered} = \llbracket \text{nil}, \text{ok cons} \rrbracket$                                                                             |
| 4 $= \text{perm}^\circ \text{ ordered}^\circ$                                         | $(R S)^\circ = S^\circ R^\circ$                                                                                                                          |
| 5 $= \text{perm ordered}^\circ$                                                       | $\text{perm}^\circ = \text{perm}$                                                                                                                        |
| 6 $= \text{perm ordered}$                                                             | $\text{ordered}^\circ = \text{ordered}$                                                                                                                  |

 **$\text{select}^\circ \sqsubseteq \text{ok cons perm} \implies (1 \times \text{perm}) \text{ select}^\circ \sqsubseteq \text{ok cons perm}$** 

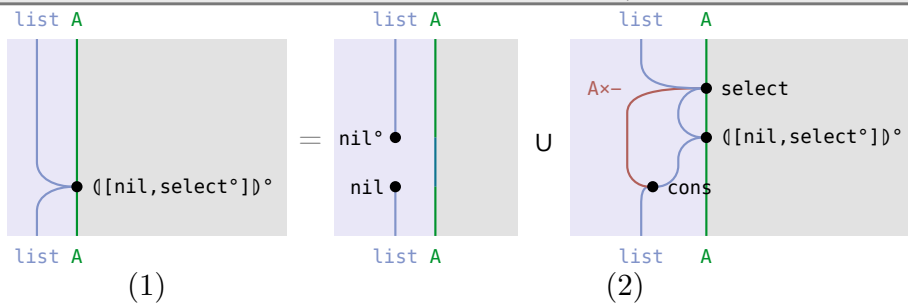
(6.6.2b)

permuting the tail and then undoing **select** lands among the **ok** conses of a permutation

|                                                             |                                                                         |
|-------------------------------------------------------------|-------------------------------------------------------------------------|
| 1 $(1 \times \text{perm}) \text{ select}^\circ$             | $\text{select}^\circ \sqsubseteq \text{ok cons perm}$                   |
| 2 $\sqsubseteq (1 \times \text{perm}) \text{ ok cons perm}$ | $\text{ok } (1 \times \text{perm}) = (1 \times \text{perm}) \text{ ok}$ |
| 3 $= \text{ok } (1 \times \text{perm}) \text{ cons perm}$   | $\text{cons perm} = (1 \times \text{perm}) \text{ cons perm}$           |
| 4 $= \text{ok cons perm}$                                   |                                                                         |

$$([nil, select^\circ])^\circ = (nil^\circ nil) \cup (select (1 \times ([nil, select^\circ])^\circ) cons) \quad (6.6.2c)$$

selection sort returns [] on [], and otherwise selects (a,y), sorts y and conses a back on



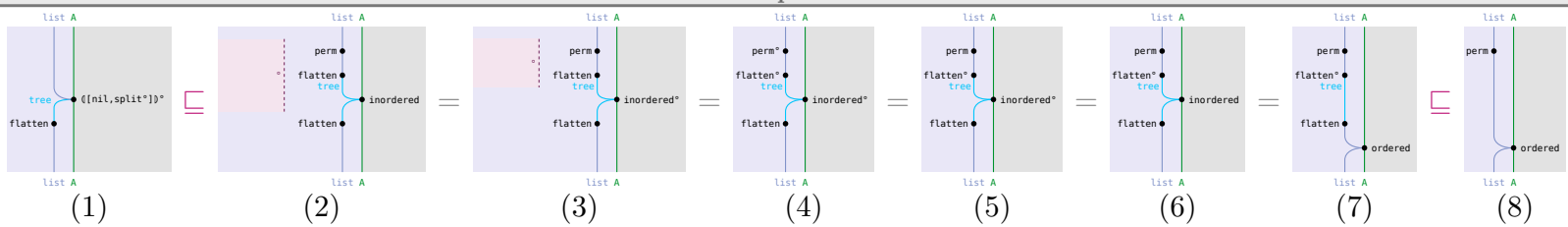
|   |                                                                               |                                  |
|---|-------------------------------------------------------------------------------|----------------------------------|
| 1 | $[nil, select^\circ]^\circ$                                                   | unfolding the converse of a fold |
| 2 | $= (nil^\circ nil) \cup (select (1 \times ([nil, select^\circ])^\circ) cons)$ |                                  |

### §6.6.3. Quicksort

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge split^\circ \sqsubseteq check' \text{ join perm} \Rightarrow ([nil, split^\circ])^\circ \quad (6.6.3a)$$

**flatten  $\sqsubseteq$  perm ordered**

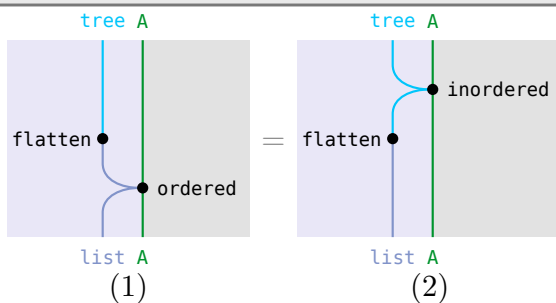
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



|   |                                                                      |                                                                                                                                                                                   |
|---|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $[nil, split^\circ]^\circ \text{ flatten}$                           | $split^\circ \sqsubseteq check' \text{ join perm}$<br>$\Rightarrow ([nil, split^\circ])^\circ \text{ flatten} \sqsubseteq (inordered \text{ flatten perm})^\circ \text{ flatten}$ |
| 2 | $\sqsubseteq (inordered \text{ flatten perm})^\circ \text{ flatten}$ | $(R S)^\circ = S^\circ R^\circ$                                                                                                                                                   |
| 3 | $= (flatten \text{ perm})^\circ inordered^\circ \text{ flatten}$     | $(R S)^\circ = S^\circ R^\circ$                                                                                                                                                   |
| 4 | $= perm^\circ flatten^\circ inordered^\circ \text{ flatten}$         | $perm^\circ = perm$                                                                                                                                                               |
| 5 | $= perm \text{ flatten}^\circ inordered^\circ \text{ flatten}$       | $inordered^\circ = inordered$                                                                                                                                                     |
| 6 | $= perm \text{ flatten}^\circ inordered \text{ flatten}$             | (6.6.3b) $(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c))$<br>$\Rightarrow \text{flatten ordered} = inordered \text{ flatten}$                    |
| 7 | $= perm \text{ flatten}^\circ \text{ flatten ordered}$               | $flatten^\circ \text{ flatten} \sqsubseteq 1$                                                                                                                                     |
| 8 | $\sqsubseteq perm \text{ ordered}$                                   |                                                                                                                                                                                   |

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = inordered \text{ flatten} \quad (6.6.3b)$$

flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening

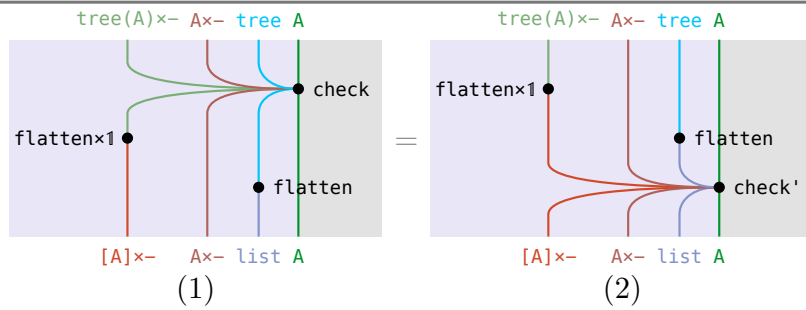


|   |                               |                                                                       |
|---|-------------------------------|-----------------------------------------------------------------------|
| 1 | <b>flatten ordered</b>        | R transitive: a flattened tree is ordered iff every node passes check |
| 2 | $= inordered \text{ flatten}$ |                                                                       |

**check (flatten × (1 × flatten)) = (flatten × (1 × flatten)) check'**

(6.6.3c)

testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening

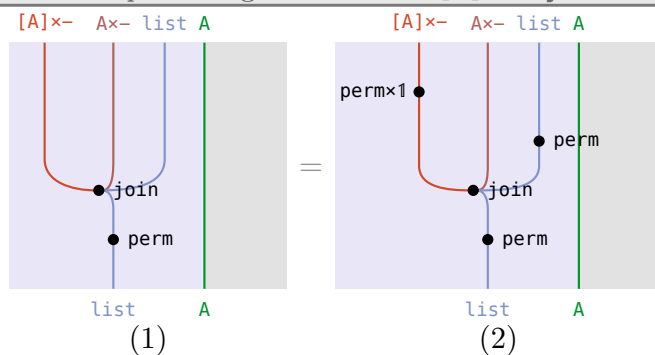


|   |                                    |                                  |
|---|------------------------------------|----------------------------------|
| 1 | check (flatten × (1 × flatten))    | b intree x ⇔ b inlist flatten(x) |
| 2 | = (flatten × (1 × flatten)) check' |                                  |

**join perm = (perm × (1 × perm)) join perm**

(6.6.3d)

permuting the list **x ++ [a] ++ y** is the same as first permuting **x** and **y** and then permuting the joined list

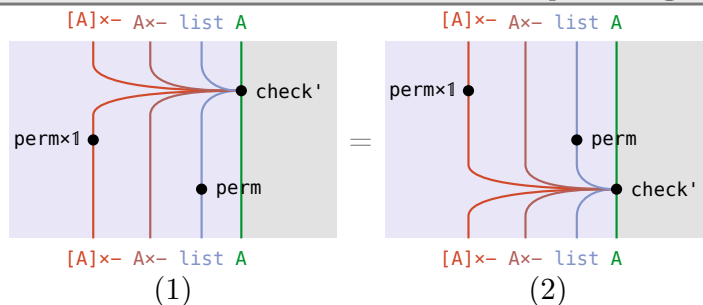


|   |                                 |                                        |
|---|---------------------------------|----------------------------------------|
| 1 | join perm                       | perm is transitive and closed under ++ |
| 2 | = (perm × (1 × perm)) join perm |                                        |

**check' (perm × (1 × perm)) = (perm × (1 × perm)) check'**

(6.6.3e)

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

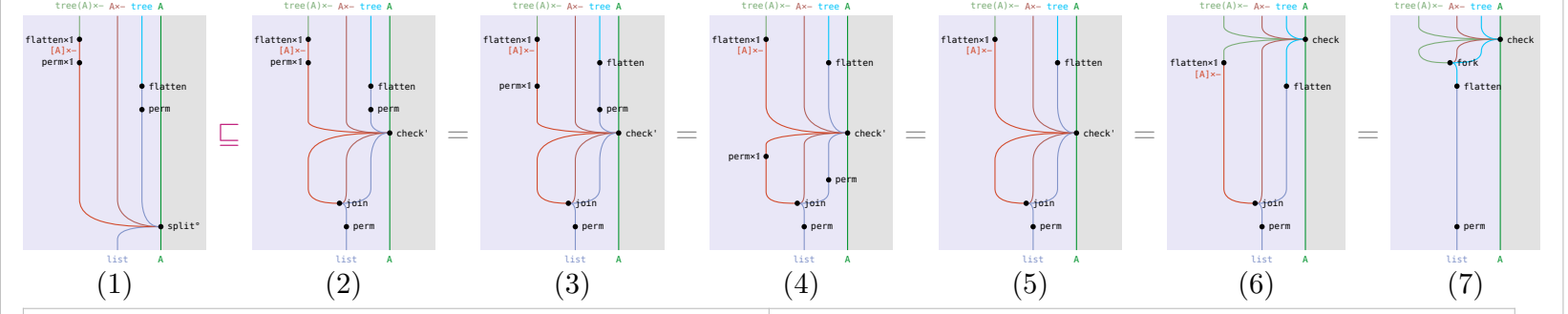


|   |                              |                               |
|---|------------------------------|-------------------------------|
| 1 | check' (perm × (1 × perm))   | b inlist x ⇔ b inlist perm(x) |
| 2 | = (perm × (1 × perm)) check' |                               |

**split° ⊆ check' join perm ⇒ ((flatten perm) × (1 × (flatten perm))) split° ⊆ check fork** (6.6.3f)

**flatten perm**

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is **permuted**

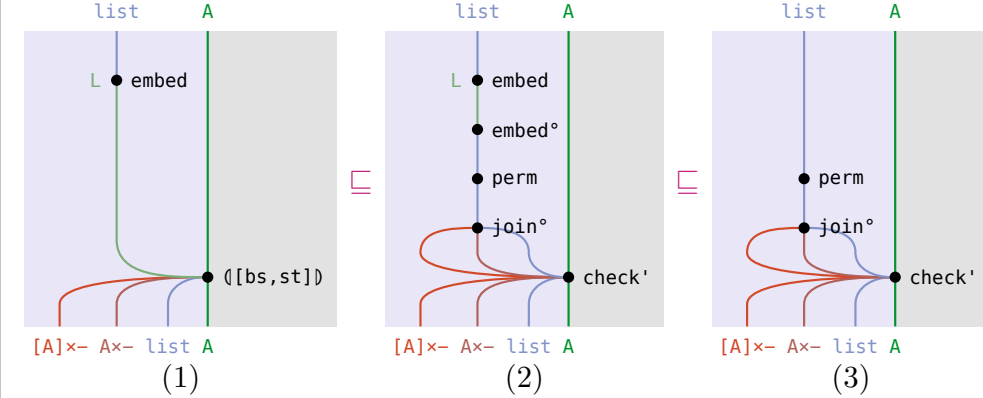


|   |                                                                                                                                   |                                                                                                                                             |
|---|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ$                                              | $\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$                                                                            |
| 2 | $\sqsubseteq ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{check}' \text{ join perm}$                     |                                                                                                                                             |
| 3 | $= (\text{flatten} \times (1 \times \text{flatten})) (\text{perm} \times (1 \times \text{perm})) \text{check}' \text{ join perm}$ | (6.6.3e) $\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{check}'$            |
| 4 | $= (\text{flatten} \times (1 \times \text{flatten})) \text{check}' (\text{perm} \times (1 \times \text{perm})) \text{join perm}$  | (6.6.3d) $\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{join perm}$                                                  |
| 5 | $= (\text{flatten} \times (1 \times \text{flatten})) \text{check}' \text{ join perm}$                                             | (6.6.3c) $\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{check}'$ |
| 6 | $= \text{check} (\text{flatten} \times (1 \times \text{flatten})) \text{join perm}$                                               | $\text{fork flatten} = (\text{flatten} \times (1 \times \text{flatten})) \text{join}$                                                       |
| 7 | $= \text{check fork flatten perm}$                                                                                                |                                                                                                                                             |

**bs ⊆ wrap perm join° check' ∧ (1 × (perm join° check')) st ⊆ cons perm join° check' ⇒** (6.6.3g)

**embed ([bs,st]) ⊆ perm join° check'**

turning a list into a non-empty list and folding it with any **[bs, st]** meeting the two conditions below gives only **checked** triples **(x,a,y)** whose join  $x \# [a] \# y$  is a permutation of the list

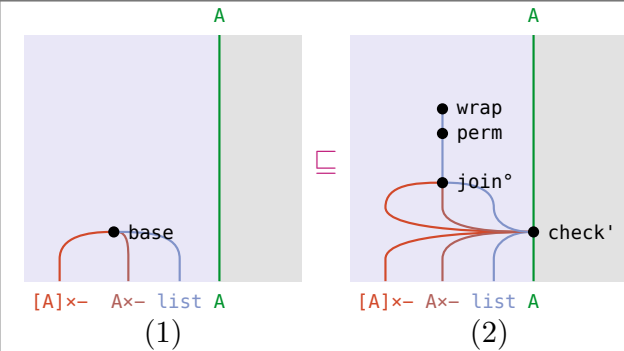


|   |                                                                             |                                                                                                                                                                                                                                                                                                               |
|---|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\text{embed } ([bs,st])$                                                   | $\text{bs} \sqsubseteq \text{wrap perm join}^\circ \text{check}' \wedge (1 \times (\text{perm join}^\circ \text{check}')) \text{st} \sqsubseteq \text{cons perm join}^\circ \text{check}'$<br>$\Rightarrow \text{embed } ([bs,st]) \sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$ |
| 2 | $\sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$ | $\text{embed embed}^\circ \text{perm join}^\circ \text{check}' \sqsubseteq \text{perm join}^\circ \text{check}'$                                                                                                                                                                                              |
| 3 | $\sqsubseteq \text{perm join}^\circ \text{check}'$                          |                                                                                                                                                                                                                                                                                                               |

**base ⊆ wrap perm join° check'**

(6.6.3h)

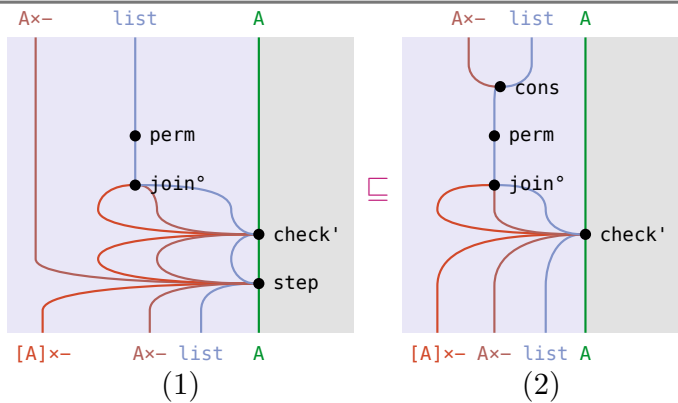
**base(a)** is a **checked** triple whose join is a permutation of the one-element list **[a]**



|   |                                                         |                                                                        |
|---|---------------------------------------------------------|------------------------------------------------------------------------|
| 1 | <b>base</b>                                             | $[\ ] \# [a] \# [\ ] = [a]$ , and $\text{check}'$ holds on empty lists |
| 2 | $\sqsubseteq \text{wrap perm join}^\circ \text{check}'$ |                                                                        |

**connected(R)  $\Rightarrow$  (1  $\times$  (perm join $^\circ$  check')) step  $\sqsubseteq$  cons perm join $^\circ$  check'** (6.6.3i)

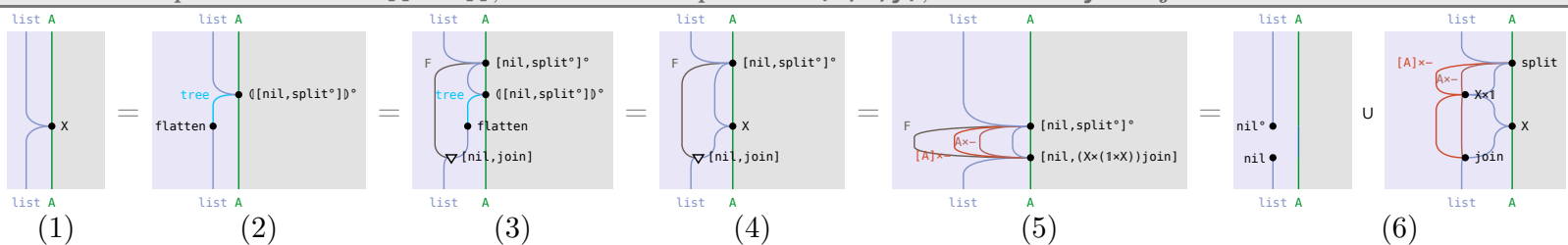
if (x,b,y) is a **checked** triple for a permutation of **l**, then **step(a, (x,b,y))** is a **checked** triple for a permutation of **[a] # l**



|   |                                               |                                                                  |
|---|-----------------------------------------------|------------------------------------------------------------------|
| 1 | (1 $\times$ (perm join $^\circ$ check')) step | aRb puts a in front of x; otherwise bRa and a goes in front of y |
| 2 | $\sqsubseteq$ cons perm join $^\circ$ check'  |                                                                  |

**X = ([nil,split $^\circ$ ]) $^\circ$  flatten  $\Rightarrow$  X = (nil $^\circ$  nil) u (split (X  $\times$  (1  $\times$  X)) join)** (6.6.3j)

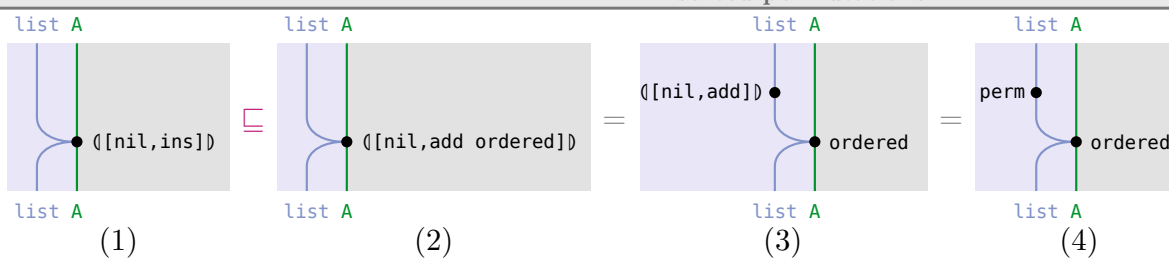
quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a



|   |                                                                                       |                                                        |
|---|---------------------------------------------------------------------------------------|--------------------------------------------------------|
| 1 | X                                                                                     | X = ([nil,split $^\circ$ ]) $^\circ$ flatten           |
| 2 | = ([nil,split $^\circ$ ]) $^\circ$ flatten                                            | S $^\circ$ F((S) $^\circ$ (R)) R = (S) $^\circ$ (R)    |
| 3 | = [nil,split $^\circ$ ] $^\circ$ F([nil,split $^\circ$ ] $^\circ$ flatten) [nil,join] | X = ([nil,split $^\circ$ ]) $^\circ$ flatten           |
| 4 | = [nil,split $^\circ$ ] $^\circ$ F(X) [nil,join]                                      | F(S) [T,U] = [T,(S $\times$ (1 $\times$ S)) U]         |
| 5 | = [nil,split $^\circ$ ] $^\circ$ [nil,(X $\times$ (1 $\times$ X)) join]               | [U,V] $^\circ$ [R,S] = (U $^\circ$ R) u (V $^\circ$ S) |
| 6 | = (nil $^\circ$ nil) u (split (X $\times$ (1 $\times$ X)) join)                       |                                                        |

**(1  $\times$  ordered) ins  $\sqsubseteq$  add ordered  $\Rightarrow$  ([nil,ins])  $\sqsubseteq$  perm ordered** (6.6.3k)

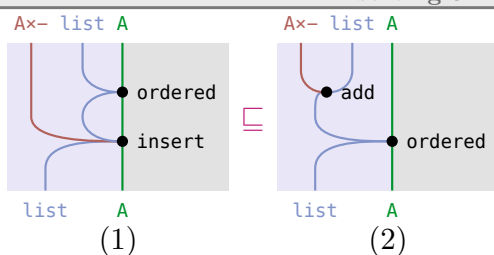
for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations



|   |                                 |                                                             |
|---|---------------------------------|-------------------------------------------------------------|
| 1 | [nil,ins]                       | (1 $\times$ ordered) ins $\sqsubseteq$ add ordered          |
| 2 | $\sqsubseteq$ [nil,add ordered] | $\Rightarrow$ ([nil,ins]) $\sqsubseteq$ ([nil,add ordered]) |
| 3 | = [nil,add] ordered             | [nil,add] ordered = [nil,add ordered]                       |
| 4 | = perm ordered                  | perm = [nil,add]                                            |

**connected(R)  $\wedge$  ( $\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)$ )  $\Rightarrow$  (1  $\times$  ordered) insert  $\sqsubseteq$  add ordered** (6.6.3l)

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



|   |                             |                                                     |
|---|-----------------------------|-----------------------------------------------------|
| 1 | (1 $\times$ ordered) insert | insert splices a in, and R transitive and connected |
| 2 | $\sqsubseteq$ add ordered   |                                                     |

## §6.7. Closure

$$R^* = (\mu X : 1 \cup (X R)) \quad (6.7a)$$

closing  $R$  by composing it on the left and closing it by composing on the right give the same relation  $R^*$

$$1 \in X \wedge X X \subseteq X \implies R \subseteq X \iff R^* \subseteq X \quad (6.7b)$$

$X$  contains  $R$  exactly when it contains  $R^*$

$$1 \in R^* \quad (6.7c)$$

$R^*$  is reflexive

|   |                              |                        |
|---|------------------------------|------------------------|
| 1 | $1$                          | $R \subseteq R \cup S$ |
| 2 | $\sqsubseteq 1 \cup (R R^*)$ | $1 \cup (R R^*) = R^*$ |
| 3 | $= R^*$                      |                        |

$$R \subseteq R^* \quad (6.7d)$$

$R^*$  contains  $R$

|   |                     |                       |
|---|---------------------|-----------------------|
| 1 | $R$                 |                       |
| 2 | $= R$               | $(6.7c) 1 \in R^*$    |
| 3 | $\sqsubseteq R R^*$ | $R R^* \subseteq R^*$ |
| 4 | $\sqsubseteq R^*$   |                       |

$$(1 \cup (R (R^* / R^*))) R^* \subseteq R^* \quad (6.7e)$$

$R^*/R^*$  is a prefixed point of  $X \mapsto 1 \cup RX$ , stated through division

$$R^* R^* \subseteq R^*$$

$R^*$  is transitive, because  $R^*/R^*$  is a prefixed point of  $X \mapsto 1 \cup RX$

|   |                                    |                                   |
|---|------------------------------------|-----------------------------------|
| 1 | $(1 \cup (R (R^* / R^*))) R^*$     | $(S \cup T) R = (S R) \cup (T R)$ |
| 2 | $= (R^*) \cup (R (R^* / R^*) R^*)$ |                                   |
| 3 | $= R^* \cup (R (R^* / R^*) R^*)$   | $(R / R) R \subseteq R$           |
| 4 | $\sqsubseteq R^* \cup (R R^*)$     | $R R^* \subseteq R^*$             |
| 5 | $\sqsubseteq R^*$                  |                                   |

$$1 \in X \wedge X X \subseteq X \wedge R \subseteq X \implies 1 \cup (R X) \subseteq X \quad (6.7f)$$

a reflexive transitive  $X$  containing  $R$  is a prefixed point of  $Y \mapsto 1 \cup RY$

$$1 \in X \wedge X X \subseteq X \wedge R \subseteq X \implies R^* \subseteq X$$

a reflexive transitive  $X$  containing  $R$  is a prefixed point of  $X \mapsto 1 \cup RX$ , so it contains  $R^*$

|   |                            |                 |
|---|----------------------------|-----------------|
| 1 | $1 \cup (R X)$             | $1 \in X$       |
| 2 | $\sqsubseteq X \cup (R X)$ | $R \subseteq X$ |
| 3 | $\sqsubseteq X \cup (X X)$ |                 |
| 4 | $\sqsubseteq X$            |                 |

$$\frac{R^*}{\exists} = \left( \frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup} \quad (6.7g)$$

the set of  $R^*$ -successors of  $a$  is  $a$  itself joined with the  $R^*$ -successors of its  $R$ -successors

Diagram illustrating the construction of  $R^*$ -successors in three steps:

- (1) A node  $a$  with a self-loop labeled  $1$  and an outgoing edge labeled  $R$  to a node  $R^*$ .
- (2) The same node  $a$  with a self-loop labeled  $1$  and an outgoing edge labeled  $R R^*$  to a node  $R R^*$ .
- (3) The result of the cup operation: a node  $a$  with a self-loop labeled  $1$  and an outgoing edge labeled  $R$  to a node  $R R^*$ , which then has a self-loop labeled  $R R^*$  to itself. The entire structure is labeled  $\text{cup}$ .

|   |                                                                         |                                                                                              |
|---|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 1 | $\frac{R^*}{\exists}$                                                   | $1 \cup (R R^*) = R^*$                                                                       |
| 2 | $= \frac{1 \cup (R R^*)}{\exists}$                                      | $\frac{R \cup S}{\exists} = \left( \frac{R}{\exists}, \frac{S}{\exists} \right) \text{ cup}$ |
| 3 | $= \left( \frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup}$ |                                                                                              |

## §6.7.1. Computing closure

|                                                             |          |
|-------------------------------------------------------------|----------|
| $R - \emptyset = R$                                         | (6.7.1a) |
| taking nothing away leaves $R$                              |          |
| $R \cup (S - R) = R \cup S$                                 |          |
| beside $R$ , only the part of $S$ outside $R$ adds anything |          |
| $R - (S \cup T) = R - S - T$                                |          |
| taking away $S$ and then $T$ takes away $S \cup T$          |          |
| $(R \cup S) - T = (R - T) \cup (S - T)$                     |          |
| taking $T$ away from a union takes it away from each part   |          |

|                                                                                                                                   |          |
|-----------------------------------------------------------------------------------------------------------------------------------|----------|
| $\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$ | (6.7.1b) |
| the least fixed point of $\varphi$ after $\psi$ is $\varphi$ applied to the least fixed point of $\psi$ after $\varphi$           |          |

|                                                                                                               |          |
|---------------------------------------------------------------------------------------------------------------|----------|
| $S R^* = (\mu X : S \cup (X R))$                                                                              | (6.7.1c) |
| $S$ followed by any number of $R$ steps is the least $X$ containing $S$ and closed under a further $R$ step   |          |
| $R^* S = (\mu X : S \cup (R X))$                                                                              |          |
| any number of $R$ steps followed by $S$ is the least $X$ containing $S$ and closed under an $R$ step in front |          |

|                                                                          |                                                               |
|--------------------------------------------------------------------------|---------------------------------------------------------------|
| $\theta(\emptyset, S) = S R^*$                                           | (6.7.1d)                                                      |
| started with nothing found and $S$ to explore, $\theta$ computes $S R^*$ |                                                               |
| 1 $\theta(\emptyset, S)$                                                 | $\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$ |
| 2 $\triangleq \emptyset \cup (\mu X : S \cup ((X R) - \emptyset))$       | (6.7.1a) $R - \emptyset = R$                                  |
| 3 $= \emptyset \cup (\mu X : S \cup (X R))$                              | $\emptyset \cup R = R$                                        |
| 4 $= (\mu X : S \cup (X R))$                                             | (6.7.1c) $S R^* = (\mu X : S \cup (X R))$                     |
| 5 $= S R^*$                                                              |                                                               |

|                                                                  |                                                               |
|------------------------------------------------------------------|---------------------------------------------------------------|
| $\theta(P, \emptyset) = P$                                       | (6.7.1e)                                                      |
| with nothing left to explore, $\theta$ returns what it has found |                                                               |
| 1 $\theta(P, \emptyset)$                                         | $\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$ |
| 2 $\triangleq P \cup (\mu X : \emptyset \cup ((X R) - P))$       | $(\mu X : \emptyset \cup ((X R) - P)) = \emptyset$            |
| 3 $= P \cup \emptyset$                                           | $R \cup \emptyset = R$                                        |
| 4 $= P$                                                          |                                                               |